OPERATING SYSTEMS (CS402) | SEMESTER 4

Operating Systems — Virtual Memory, Page Replacement & Thrashing Prevention Algorithms

Topic / Unit: Memory Management Systems & Hardware Translation

Contributed by: Aarav Sharma (B.Tech CSE Year 2) | Upload Date: 2026-10-23

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1. EXECUTIVE SUMMARY & OVERVIEW

Virtual memory is a storage allocation scheme in which secondary memory can be addressed as though it were part of main memory. The addresses a program may use to reference memory are distinguished from the addresses the memory system uses to identify physical storage sites, and program-generated addresses are translated automatically to the corresponding machine addresses. This unit provides comprehensive lecture notes covering demand paging, effective access time calculations using TLB hit rates, inverted page tables, segmentation with paging, and FIFO, Optimal, and LRU page replacement algorithm evaluations with complete numerical traces.

2. KEY CONCEPTS & CORE BLUEPRINT

[1] Demand Paging Architecture: Hardware-supported page fault trap mechanism, swap space management, and effective memory access time formulas with dirty bit optimizations.

[2] Page Replacement Algorithms: FIFO, Least Recently Used (LRU) with stack and counter implementations, Optimal (OPT/MIN) policy, and Clock second-chance approximation.

[3] Belady's Anomaly Proof: Mathematical demonstration that increasing physical page frame count can paradoxically increase page fault counts under FIFO queuing.

[4] Working Set Model and Thrashing: Locality of reference analysis, page fault frequency (PFF) controls, and multiprogramming degree management to stabilize system throughput.

[5] Memory Allocation Strategies: Equal vs Proportional allocation across competing high-priority kernel threads and user processes.

3. REVISION GUIDELINES & MODEL PROBLEM APPROACH

- Master core theoretical definitions and primary architectural diagrams before attempting derivation steps.
- Formulate step-by-step algorithms, verify edge cases (null pointers, boundary indices, overflow limits).
- Practice university previous-year question papers (last 3-5 iterations) mapped to this specific syllabus unit.

- Draft concise summaries of proofs, standard equations, and time/space complexity Big-O upper bounds.
- For multi-part descriptive questions, structure answers with clear headings, schematic diagrams, and brief bullet conclusions.

4. EXAMINATION PREPARATION & ANSWER STRUCTURING

- High-Yield Diagrams: Draw neat, labeled diagrams for all system architecture and flowchart questions to secure full presentation marks.
- Numerical Precision: Show every intermediate substitution step and always state explicit engineering units in final calculations.
- Time Allocation: Allocate 1.5 minutes per mark (e.g., maximum 15 minutes for a 10-mark long question) during mid-term and semester finals.

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